

KANKAANPAA NIINISALO PUOLVOIM, Finland

WMO: 029420

Lat: 61.839N

Lon: 22.463E

Elev: 409

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.5	-3.6	-13.8	2.6	-9.1	-7.7	3.7	-3.3	18.0	28.4	15.9	28.4	3.8	80	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	80.3	64.6	76.8	62.8	73.3	61.0	67.4	75.9	65.2	72.9	63.1	70.6	7.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	92.3	71.0	62.2	85.0	68.5	60.1	78.7	66.1	32.1	76.3	30.3	73.0	28.8	71.1	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	14.2	12.6	DB	-14.3	84.0	6.9	3.9	-19.2	86.9	-23.2	89.2	-27.1	91.4	-32.1	94.2	(4)
(5)			WB	-14.7	69.8	6.9	3.3	-19.7	72.2	-23.7	74.1	-27.6	75.9	-32.6	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	20.8	20.6	27.4	38.1	48.5	56.6	62.2	59.6	50.2	39.8	31.8	25.7		
	DBStd	16.59	10.98	11.58	8.38	6.38	7.71	6.40	5.78	5.64	6.14	7.21	8.04	10.69		
	HDD50	4617	904	824	701	360	121	12	0	2	72	320	546	755		
	HDD65	9111	1369	1244	1166	807	514	264	125	183	443	780	996	1220		
	CDD50	1052	0	0	0	3	75	209	380	301	79	5	0	0		
	CDD65	70	0	0	0	0	2	11	40	17	0	0	0	0		
	CDH74	648	0	0	0	0	37	129	359	123	0	0	0	0		
	CDH80	110	0	0	0	0	2	18	78	12	0	0	0	0		
	WSAvg	6.2	5.8	6.3	6.8	6.4	6.6	6.3	5.9	5.7	5.8	6.3	6.3	6.2		
	PrecAvg	24.7	1.7	1.3	1.4	1.4	1.5	2.3	3.0	3.0	2.7	2.4	2.2	1.9		
(15) Precipitation	PrecMax	34.7	3.5	3.7	2.6	3.3	3.5	5.4	6.9	7.0	5.6	4.6	5.2	3.9		
	PrecMin	18.0	0.4	0.1	0.0	0.2	0.2	0.2	0.1	0.5	0.4	0.8	0.3	0.4		
	PrecStd	4.1	0.8	0.8	0.6	0.8	0.9	1.3	1.5	1.5	1.3	1.1	1.1	0.8		
	PrecStd	4.1	0.8	0.8	0.6	0.8	0.9	1.3	1.5	1.5	1.3	1.1	1.1	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	40.5	50.3	64.7	78.4	82.3	86.3	81.5	69.6	56.9	49.4	44.4		
		MCWB	38.8	37.2	40.8	50.7	59.3	63.6	68.0	66.1	60.5	52.1	48.2	43.4		
	2%	DB	37.0	37.5	44.4	58.9	73.2	77.7	81.8	77.4	66.2	53.9	46.4	41.7		
		MCWB	36.1	35.9	38.2	47.8	57.1	61.7	66.2	64.3	58.4	51.7	45.3	40.9		
	5%	DB	35.2	35.4	40.7	54.4	68.2	73.4	78.0	73.8	62.8	52.1	44.2	39.5		
		MCWB	34.5	34.4	35.9	44.7	54.6	59.2	64.3	62.3	56.3	50.2	43.4	38.7		
	10%	DB	33.3	33.5	37.6	50.2	62.9	69.6	74.0	70.5	59.9	50.1	42.2	37.4		
		MCWB	32.8	32.8	34.2	42.4	51.3	57.4	62.6	60.9	55.4	48.5	41.5	36.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.1	38.1	42.3	52.4	62.0	67.1	71.0	69.9	62.0	53.9	48.4	43.7		
		MCDB	39.4	39.3	48.8	61.8	73.3	77.5	81.5	77.3	67.3	55.1	49.3	44.0		
	2%	WB	36.3	36.1	39.0	48.7	58.8	63.7	68.4	66.4	59.7	52.2	45.6	41.2		
		MCDB	36.7	37.2	43.0	56.8	69.3	73.1	77.6	73.5	63.9	53.4	46.2	41.7		
	5%	WB	34.7	34.6	36.7	45.8	55.7	61.1	66.3	64.2	57.7	50.5	43.5	38.8		
		MCDB	35.1	35.5	39.8	52.5	66.1	70.0	74.2	71.3	61.4	51.7	44.0	39.3		
	10%	WB	32.9	32.9	34.9	43.2	52.8	59.0	64.2	62.2	56.0	48.4	41.6	36.7		
		MCDB	33.3	33.5	37.1	49.6	61.4	67.7	71.9	68.4	59.3	49.7	42.1	37.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.5	13.9	15.9	18.6	17.6	16.9	16.2	14.1	9.7	7.0	8.2		
		MCDBR	7.3	8.7	16.9	24.0	25.5	24.3	23.1	21.2	17.8	10.9	7.5	7.5		
		MCWBR	7.3	7.7	11.2	14.1	12.6	11.5	10.3	10.1	10.7	8.7	7.3	7.2		
		MCDBR	7.1	8.2	14.3	21.4	22.8	20.7	19.4	18.0	14.8	9.4	7.3	6.9		
	5% WB	MCWBR	7.2	7.5	10.2	13.2	12.1	10.9	10.0	10.0	9.5	8.4	7.1	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.298	0.321	0.360	0.356	0.351	0.376	0.362	0.338	0.320	0.270	0.351		
	taud		2.104	2.207	2.295	2.343	2.424	2.464	2.413	2.445	2.504	2.404	2.178	2.577		
	Ebn at Noon		140	209	250	263	274	278	266	260	247	204	130	89		
	Edh at Noon		11	21	27	32	32	31	32	29	23	17	10	7		
(44) All-Sky Solar Radiation	RadAvg		65	244	716	1169	1589	1704	1600	1244	741	308	79	35		
	RadStd		11	55	84	138	167	135	159	134	90	61	15	4		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page